SAN DIEGO SERVICE OPERATIONS

MASTER CASE STUDY

Business Analysis Case Study

From operational analytics to an end-to-end BA portfolio engagement

Portfolio disclosure. This is a simulated business-analysis engagement built from public City of San Diego Get It Done data and a completed Power BI analytics project. It does not represent City employment, internal-system access, stakeholder interviews, approved requirements, or formal City acceptance.

Prepared by Chase Brangham

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Executive Summary

Business problem. Get It Done service operations generate a large volume of public service-request records across multiple service categories and council districts. The business need is not simply “a dashboard”; it is a consistent way to understand backlog size, aging, workload concentration, throughput, duplicate reporting, referral activity, and where additional operational investigation may be warranted.

Approach. The companion Power BI analytics project was completed first. This case study retrospectively applies Business Analyst practices around that implemented solution: stakeholder analysis, current-state modeling, gap analysis, requirements, user stories, prioritization, KPI governance, business/data rules, traceability, UAT design, solution verification, RAID, implementation planning, and reflection.

800K	86K	75.1%	367	13.1%	6.0%
Total Requests	Open Backlog	90+ Day Backlog Rate	Median Open Case Age	Duplicate Rate	Referral Rate

Snapshot metrics from the completed companion analytics project. They are descriptive findings from the analyzed data, not City performance targets.

Primary conclusion. The portfolio evidence supports an analytics-enabled monitoring and prioritization solution, but it does not support causal claims about staffing, service quality, or why specific districts/categories have larger backlogs. Those questions require internal operational data and stakeholder/process-owner validation.

1. Business Context & Problem Definition

1.1 Business context

The City of San Diego Get It Done program receives and routes reports for non-emergency problems. The public dataset includes request attributes and status information, but it explicitly does not include details of the work performed to fix a problem or a definitive physical work-completion timestamp. That evidence boundary shapes the entire case study.

1.2 Problem statement

Operational reviewers need to understand where unresolved requests are accumulating, how old the open workload is, which service categories or geographic areas are most concentrated, and whether observed intake and dispositions are moving in balance. Without a consolidated and governed analytical view, identifying priorities and communicating trends becomes more difficult.

1.3 Scope boundaries

- In scope: public Get It Done data used in the companion analytics project; backlog/aging; service category; council district; intake/disposition trends; duplicate/referral patterns; BA documentation and traceability.
- Out of scope: internal City staffing, budgets, detailed work-order procedures, production implementation, approved SLAs, physical resolution verification, or claims of causal root cause.
- The 90-day aging threshold is an analytical monitoring threshold used by the portfolio, not an asserted City-wide SLA.

2. BA Approach & Deliverable Strategy

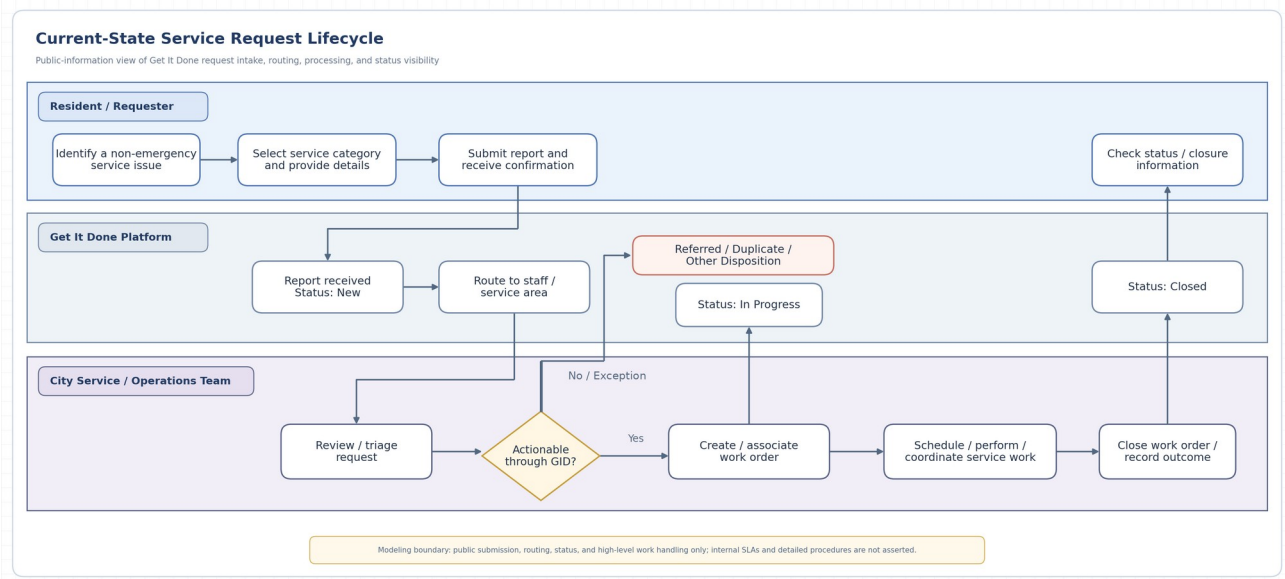
The case study uses a two-layer structure. The recruiter-facing layer explains the business story from problem to recommendation. The artifact layer provides the detailed work products a BA would use to support the story.

Stage	BA Work	Purpose
1	Problem & Scope	Define the business problem before proposing the solution.
2	Stakeholders	Identify who needs what information and how they would be engaged.
3	Current State	Model the public request lifecycle and document observed operational signals.
4	Gap Analysis	Translate observed conditions into needs/capabilities.
5	Requirements	Define business, functional, and non-functional requirements.
6	Agile Translation	Express selected requirements as user stories and acceptance criteria.
7	Governance	Define KPIs, fields, business rules, data rules, and priorities.
8	Future State	Show the proposed analytics-supported monitoring layer.
9	Traceability & Testing	Link requirements to capabilities, verification, and planned UAT.
10	Implementation & Reflection	Plan adoption/change and identify what would differ in a live engagement.

Because the dashboard existed before this BA extension, requirements are explicitly labeled Reconstructed or Proposed rather than presented as stakeholder-approved pre-build requirements.

3. Current-State Analysis

3.1 Current-State Process Map



The process map is intentionally high-level and public-information-based. Internal departmental work-order procedures and SLAs are not asserted.

3.2 Observed operational signals

- Approximately 86K requests remain open in the analyzed snapshot; about 65K are aged 90+ days, producing a 75.1% 90+ day backlog rate.
- Median open-request age is approximately 367 days.
- Sidewalk Repair Issues and Street Light Maintenance each contribute about 14.8K open requests; ROW Maintenance contributes about 10.6K.
- Council District 3 has the largest observed open backlog at roughly 19.2K requests.
- Duplicate reporting is approximately 13.1% overall; Street Light Maintenance is about 41.3% and Encampment about 28.6% among high-volume categories.
- Referral activity is approximately 6.0%. The public documentation states referred reports can leave Get It Done visibility.
- Monthly intake often exceeds monthly dispositions, while some periods show the reverse, making Net Request Flow useful for directional monitoring.

3.3 Gap analysis

Current-State Gap	Desired Capability	Traceability
Fragmented analytical sources	Consolidated reporting model	FR-01
Aging severity not obvious from total volume	Aging KPIs and age buckets	BR-02 / FR-03
Category/geographic concentrations require grouping	Service type and district comparisons	BR-03 / BR-04
Static backlog does not show direction	Requests vs dispositions + Net Flow	BR-05

Duplicate/referral semantics complicate interpretation	Duplicate/Referral KPIs + definitions	BR-09
Metric assumptions can be misread	KPI dictionary and business rules	BR-08
Internal work process unavailable publicly	Internal elicitation/process discovery	Future live engagement

3.4 Root-cause hypotheses

The current data supports prioritization questions, not final causal answers. Examples of hypotheses requiring validation include whether severe aging reflects long-cycle infrastructure processes, whether visible unresolved conditions contribute to repeat reporting, and whether district backlog differences reflect service mix or asset density. Artifact 12 records the evidence required to test each hypothesis.

4. Stakeholder Analysis

The portfolio uses proposed stakeholder groups to demonstrate method without claiming direct City interviews. Key groups include operations leadership, department/program managers, operations/data analysts, council district staff, intake/routing staff, service-delivery teams, IT/data support, and residents/requesters.

Stakeholder Group	Primary Information Need	Power / Interest	Engagement
Operations leadership	Backlog severity, trend direction, priorities	High / High	Manage closely
Department/program managers	Service-specific workload and aging	High / High	Manage closely
Operations/data analysts	Metric logic, drill-down, data quality	Medium / High	Collaborate frequently
Council district staff	Geographic concentration/context	Medium / High	Keep informed / validate
IT/data support	Refresh, model, source integrity	High / Medium	Keep satisfied / engage on changes
Residents/requesters	Status clarity and service experience	Low formal power / High interest	Inform / research where appropriate

In a live engagement, stakeholder identification would be validated first, followed by interviews/workshops to confirm decision rights, process ownership, service-specific targets, metric definitions, and reporting cadence.

5. Requirements, Stories & Prioritization

5.1 Business requirements

ID	Business Requirement	Priority
BR-01	Consolidated operational view	Must
BR-02	Aged backlog visibility	Must
BR-03	Council district comparison	Must
BR-04	Service-category comparison	Must
BR-05	Intake vs disposition monitoring	Must
BR-06	Interactive segmentation	Should
BR-07	Investigation without unsupported causal claims	Should
BR-08	Metric definitions, assumptions, limitations	Must
BR-09	Duplicate and referral monitoring	Should

5.2 Prioritization method

Artifact 15 supplements MoSCoW with a weighted score across business value, decision criticality, risk/integrity, dependency/enablement, and inverse effort. The purpose is not to override stakeholder judgment; it is to make priority reasoning explicit and challengeable.

5.3 Agile translation

Eleven user stories translate the requirements into implementable work. Acceptance criteria are written in Given/When/Then style and map forward to UAT or documentation checks. Example: BR-02 → FR-03 → US-02 → Backlog ≥90 Days KPI → UAT-02.

6. KPI, Data & Business-Rule Governance

6.1 KPI dictionary

Metric	Business Meaning	Governance Caveat
Total Requests	Overall modeled request volume	Includes duplicate submissions unless filtered.
Open Backlog	Active/open request count	Public status is not detailed work-order stage.
90+ Day Backlog / Rate	Long-open workload count/share	90 days is an analytical threshold, not an SLA.
Median Open Case Age	Typical age of active requests	Must be tied to snapshot/refresh date.

Requests Received	Requests by date_requested	Intake measure.
Requests Dispositioned	Inactive records by date_closed	Closed/Referred does not prove physical completion.
Net Request Flow	Received − Dispositioned	Directional flow, not a full productivity measure.
Duplicate Rate	Duplicates / Total Requests	Interpretation requires process/customer context.
Referral Rate	Referred / Total Requests	Workflow exit; downstream resolution may be invisible.

6.2 Data quality lesson from implementation

During the original analytics build, approximately 800 records initially produced date-conversion errors. Investigation showed the dates were not the root problem: quoted multiline text in free-text fields caused CSV parsing to shift values across columns. The ingestion logic was corrected rather than dropping those records downstream. Artifact 14 converts that lesson into repeatable data-quality rules for refresh, status-domain checks, source row counts, parent-ID logic, and metric reconciliation.

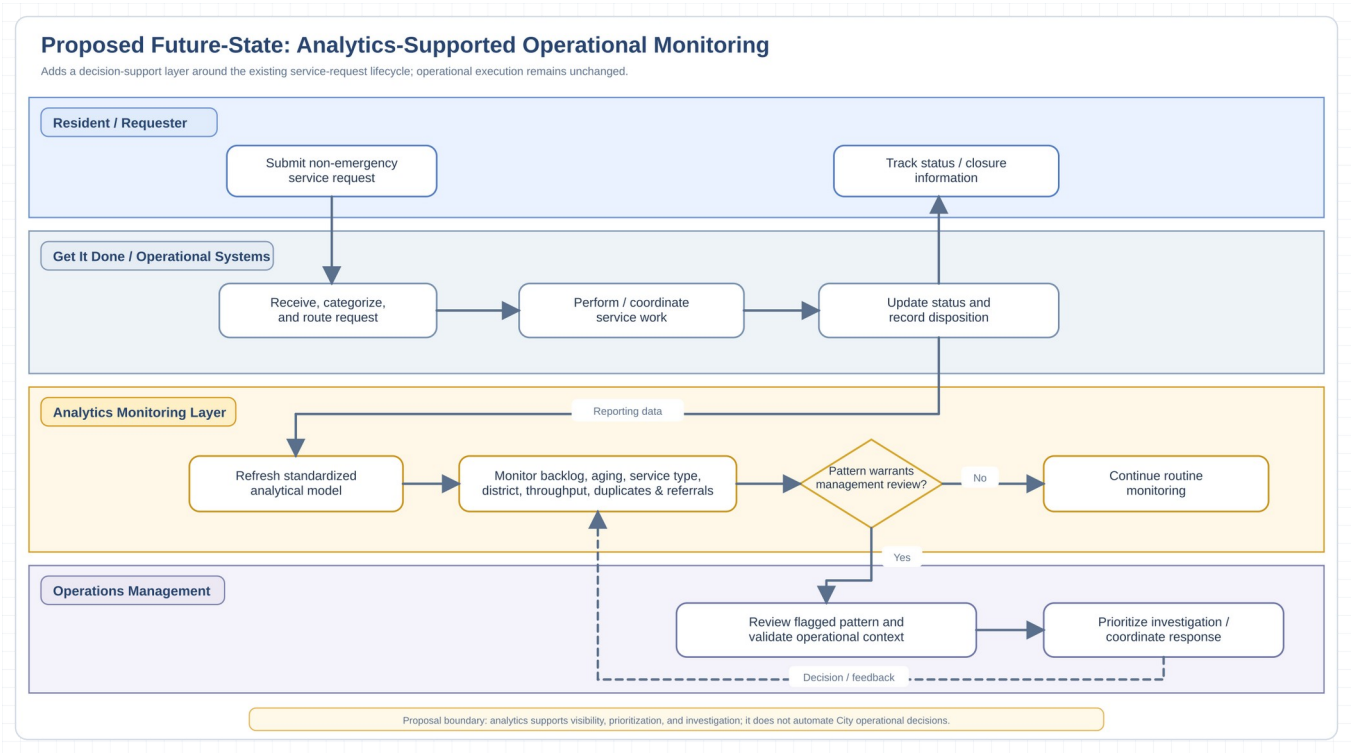
6.3 Key interpretation rules

- New and In Process are treated as active/open; Closed and Referred as inactive/workflow-exit for this case study.
- Referred is not presented as confirmed physical resolution.
- Duplicate identification uses the documented parent-request relationship.
- District volume is descriptive unless normalized/contextualized.
- Recommendations from public data are framed as investigation priorities rather than causal conclusions.

7. Proposed Future State

The proposed future state does not automate operational decisions. It creates a governed monitoring layer that refreshes standardized reporting data, surfaces backlog/aging/category/district/throughput patterns, flags conditions for management review, and records whether further analysis or operational action is needed.

The analytics monitoring layer is a portfolio design concept around the completed Power BI solution, not a claim about the City's current internal workflow.



8. Implemented Solution Evidence

The implemented analytical solution is the public companion repository: github.com/ChaseBrangham/san-diego-service-operations

8.1 Executive Overview

The first Power BI page provides leadership-level visibility into Total Requests, Open Backlog, 30+ and 90+ day backlog, median open age, service-type backlog, service-type aging rates, and monthly Requests Received vs. Requests Dispositioned.

View actual dashboard screenshot: Executive Overview

8.2 Backlog & Process Analysis

The second Power BI page provides age-bucket distribution, monthly Net Request Flow, council-district backlog, Duplicate Rate, Referral Rate, and duplicate concentration among high-volume services.

View actual dashboard screenshot: Backlog & Process Analysis

9. Traceability, Verification & UAT

9.1 End-to-end traceability

Business Req	Functional Req	Story	Solution Evidence	Validation
BR-02	FR-03	US-02	Backlog ≥90 Days KPI	UAT-02
BR-03	FR-04	US-03	Council District backlog visual	UAT-03
BR-05	FR-06 / FR-07	US-05 / US-06	Monthly trend + Net Request Flow	UAT-05 / UAT-06
BR-09	FR-13 / FR-14	US-11	Duplicate + Referral metrics	UAT-13 / UAT-14

9.2 Analyst verification

Artifact 17 separates portfolio evidence checks from formal UAT. Visible dashboard capabilities and documented implementation details can be verified from the public companion repository. Filter-interaction behavior and row-level numerical reconciliation are not treated as fully verified unless the PBIX/source model is opened and tested.

9.3 UAT posture

Fourteen business-facing UAT scenarios are defined, but they remain Planned. This is deliberate: a portfolio exercise without representative City users should not claim business acceptance. In a real engagement, UAT would include named testers, execution evidence, defect handling, retest, and formal sign-off/acceptance criteria.

10. Risks, Assumptions, Issues & Dependencies

Level	Risk / Dependency	Response
Critical	Public request data may be mistaken for completed maintenance work.	Use request/disposition terminology and explicit limitations.
Critical	Detailed internal work process is unavailable.	Bound process model to public lifecycle; require internal discovery.
High	District backlog may be misread as performance.	Require normalization/context before ranking.
High	90-day threshold may be misread as SLA.	Label as analytical threshold; validate by service.
High	Source-schema/CSV parsing changes may corrupt refresh.	Run repeatable data-quality checks and reconciliation.
Dependency	Root-cause interventions require internal process/context data.	Treat recommendations as investigation priorities until validated.

11. Recommendations & Next Steps

- Prioritize deeper operational review where high volume and severe aging overlap, particularly Sidewalk Repair, Street Light Maintenance, ROW Maintenance, and Pavement Maintenance.
- Investigate Street Light duplicate reporting as a business/process question. The observed concentration is a signal, not proof of its cause.
- Review District 3 backlog with population, service mix, asset inventory, and jurisdiction context before drawing performance conclusions.
- Monitor Requests Received vs. Requests Dispositioned and Net Request Flow over time to identify persistent workload pressure or easing.
- Treat referral as a distinct workflow outcome and, if operationally useful, standardize referral destinations for analysis.
- In a live project, validate stakeholders, service-specific metrics/targets, business rules, and detailed process maps before proposing production process changes.

12. Reflection: What This Case Study Adds

The original project proved that I could build an analytical solution. The BA extension changes the emphasis from “what does the dashboard show?” to “why does this capability exist, whose decision does it support, how is it defined, what assumptions does it depend on, and how would we know it is acceptable?”

The biggest limitation is also an important lesson: a strong BA does not fill evidence gaps with confident-sounding assumptions. Public data can show patterns, but a real engagement would require stakeholder elicitation, internal process discovery, service-level definitions, data ownership, and UAT before operational changes were approved.

If this were a live project, I would move requirements and metric governance earlier in the lifecycle, define acceptance criteria before the build, and involve process/data owners in validating status semantics and service-specific thresholds. That would reduce rework and make the final analytical solution easier to defend and maintain.

Appendix — Artifact Library

#	Artifact	Folder
01	Project Overview & Scope	docs/
02	Stakeholder Analysis	stakeholders/
03	As-Is Process Map	process-maps/
04	To-Be Process Map	process-maps/
05	Requirements Catalog	requirements/
06	User Stories & Acceptance Criteria	requirements/
07	Requirements Traceability Matrix	traceability/
08	UAT Plan & Test Cases	testing/
09	Solution Evaluation & Recommendations	evaluation/
10	Decision & Assumption Log	project-management/
11	Jira-ready Backlog Import	jira/
12	Current-State & Gap Analysis	process-analysis/
13	KPI & Data Dictionary	requirements/
14	Business Rules & Data Requirements	requirements/
15	Requirements Prioritization	requirements/
16	RAID Log	project-management/
17	Solution Verification	testing/
18	Implementation & Change Considerations	implementation/
19	Project Reflection	reflection/

Public source references

- City of San Diego Get It Done Reports dataset
- Tips for working with Get It Done data
- Get It Done status FAQ
- Companion analytics repository